That's why Newton's laws of gravitation and motion are sufficient to fly rockets to the moon without knowing the complete quantum state of the universe; our individual branch of the wave function describes an emergent almost-classical world.

Branches of the wave function, describing separate worlds, are not mentioned in the postulates of Many-Worlds. Nor are tables and chairs and air mentioned in the Core Theory of particles and forces. As the philosopher Daniel Dennett has put it, in terms that were then ported into the quantum context by David Wallace, each world is an emergent feature that captures "real patterns" within the underlying dynamics. A real pattern gives us an accurate way of talking about the world, without appealing to a comprehensive microscopic description. That's what makes emergent patterns in general, and Everettian worlds in particular, indisputably real.

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Once you believe that branches of the wave function can usefully be thought of as emergent worlds, you might start wondering why it's this set of worlds in particular. Why do we end up seeing macroscopic objects with pretty well-defined locations in space, rather than being in superpositions of different locations? Why is "space" apparently such a central concept at all? Textbooks in introductory quantum mechanics sometimes give the impression that classical behavior is inevitable once objects become very big, but that's nonsense. We have no trouble at all imagining a wave function that describes macroscopic objects in all sorts of weird superpositions. The real answer is more interesting.

We can begin to get a handle on the special nature of space by comparing how we think about position to how we think about momentum. When Isaac Newton first wrote down the equations of classical mechanics, position clearly played a privileged role, whereas velocity and momentum were derived quantities. Position is "where you are in space," while velocity is "how fast you are moving through space," and momentum is mass times velocity. Space would appear to be the main thing.

But a deeper look reveals that the concepts of position and momentum are on more of an equal footing than they first appear. Perhaps we shouldn't be surprised; after all, position and momentum are the two quantities that